Zhichen Ying

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Education

University of Warwick, PhD Student 2026.10 - 2030.10(expected)

- Advisor: Prof. Filip Rindler

Westlake University, Visiting Student 2025.09 - 2025.12

- Mentor: Prof. Zhongwei Shen
- Studied the nodal sets and critical sets in periodic homogenization and related topics.

Zhejiang University, BS in Mathematics 2022.09 - present

- GPA: 4.08/4.30, 90/100
- **Qiushi Mathematical Science Program**
- **Relevant Courseworks**: Complex Analysis(H), Real Analysis, Functional Analysis(H), PDE(H), PDE(Graduate level), Riemannian Geometry

Research

- **Research Interests:** Partial Differential Equations and Analysis, Geometric Measure Theory
- **Research Program:** National Students Research Training Program (SRTP)

  Program Name: Research on the solution to the phase transition equation in liquid crystal.

  Time: 2024.4-2025.5

  Supervisor: Prof. Wei Wang

  Program description: Studied the Landau-de Gennes model and phase transition equations in liquid crystal theory. Also studied the calculus of variations, stability and asymptotic property of minimzier.

Publications & Preprints

- Li, J., & **Ying, Z.** (2025). Lower Bound of Nodal Sets in Elliptic Homogenization and Functions with Strong Maximum Principle. arXiv preprint arXiv:2512.12305.
- Li, J., Wang, J., & **Ying, Z.** (2025). Nadirashvili's Conjecture for Elliptic PDEs and its Applications. arXiv preprint arXiv:2508.07861.

Honors & Awards

- **Mathematical CDT Studentship:** Scholarship funded by Warwick Mathmatical Institue
- **Zhejiang Provincial Government Scholarship:** Scholarship funded by Zhejiang Provincial Government, 2023 & 2024
- **Scholarship for Top-notch Talents in Basic Subjects:** Scholarship for students in Qiushi Science Class in Chu Koche Honors College, 2024 & 2025
- **The Chinese Mathematics Competitions(Undergraduate-level):** Provincial 1st Prize, 2023

Teaching

- **Teaching Assistant:**

  Ordinary Differential Equations, Spring 2025, Zhejiang University

  Partial Differential Equation (H), Fall 2025, Zhejiang University

Attended Seminars:

- **Sobolev Space and Elliptic PDEs** Summer 2024

  reference: Evans, L.C. (2010) Partial Differential Equations. 2nd Edition, Department of Mathematics, University of California, Berkeley, American Mathematical Society.

  Description: With other three undergraduate students, we studied Sobolev space, second order elliptic PDEs.
- **Introduction to Fourier Analysis** Fall 2024

  reference: Sogge, C. D. (2017). Fourier integrals in classical analysis (Vol. 210). Cambridge University Press.

  Description: With two undergraduate students and one PhD student, studied the Fourier transform, wave front sets, oscillatory integral, stationary phase estimate and non-homogeneous oscillatory integral.
- **Unique Continuation Property and Nodal Sets of Elliptic PDEs** Spring 2025

  reference: Related papers

  Description: Studied Logunov's works about nodal sets and Tolsa's work about unique continuation at the boundary. Gave several talks in a graduate seminar.
- **Geometric Measure Thoery** Fall 2025

  reference: Simon, L. (2014). Introduction to geometric measure theory. Tsinghua lectures, 2(2), 3-1.

  Description: Studied Hausdorff measure, Hausdorff dimension, BV function, area formula, rectifiable set, finite perimeter set, rectifiable varifold, basic introduction of current.

Others

Languages: Chinese(native), English(fluent)